Calepin Tutorial (PDF)

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Abstract
This document renders the *Calepin* tutorial to PDF via LaTeX. It demonstrates scholarly front matter (authors, affiliations, ORCID), the appendix, and the `include` shortcode for pulling in content from another file.

This page reviews some of the basics of Quarto and Markdown syntax supported by *Calepin*. It covers document structure, code chunks, figures, math, cross-references, citations, and other features that work out of the box.

Getting started

Calepin renders `.qmd` documents to HTML, LaTeX, Typst, and Markdown. It executes R code chunks, captures output and plots, resolves citations and cross-references, and wraps everything in a template.

To render this tutorial:

```
calepin tutorial.qmd
```

Markdown basics

Calepin supports standard CommonMark and GitHub Flavored Markdown.

Lorem ipsum dolor sit amet, *consectetur* adipiscing elit. Sed do **eiusmod tempor** incididunt ut labore.

Ordered list:

1. First item
2. Second item
3. Third item

Unordered list:

- Alpha
- Beta
- Gamma

A hyperlink and some `inline code`.

> A blockquote: Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris.

Text with a footnote[1] and another[2] .

[1] This is the first footnote.

Superscript: E = mc[2]. Strikethrough: ~~deleted~~.

En dash: – and em dash: —

Code chunks

Code chunks are fenced with triple backticks and a `{r}` header. Options go inside the chunk as `#|` pipe comments.

```r
x <- 1:10
mean(x)
```

```
[1] 5.5
```

Inline code

R expressions can be evaluated inline. The mean of 1:10 is 5.5 and pi is approximately 3.14.

Figures

Code chunks that produce plots automatically generate figures. Use `fig-cap` for a caption and a label for cross-referencing.

```r
plot(
    1:10,
    (1:10)^2,
    main = "Square Numbers",
    xlab = "x",
    ylab = "x^2"
)
```

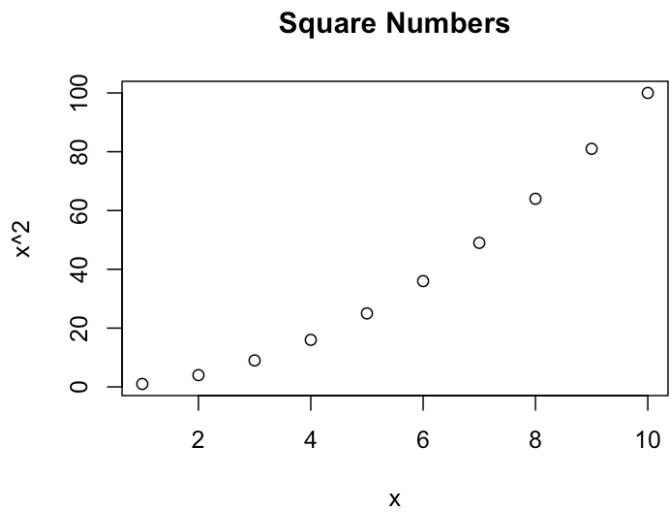

Figure 1: A simple scatter plot

[2]This is the second footnote.

Figure layouts

Figures can be arranged in grids using `layout-ncol`. The following example places three plots side by side with individual subcaptions and an overall caption.

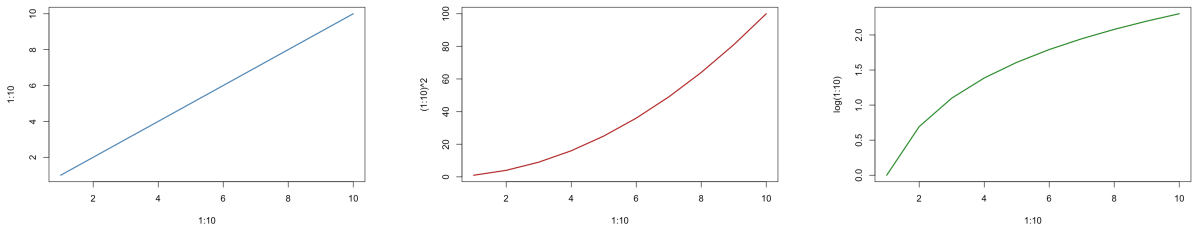

Figure 3: Linear Figure 4: Quadratic Figure 5: Logarithmic

Figure 2: Three growth patterns compared.

See Figure 5 for the result. The `layout` attribute supports custom grids with relative widths (e.g., `layout="[[1,1],[1]]"`) and negative values for spacing gaps.

Cross-references

Calepin resolves references to figures, sections, and theorem environments.

- Figure 1 renders as a linked reference
- Section 5 links to the Figures section
- 1 suppresses the type prefix

Sections get automatic ids from their heading text.

Math

We can represent dollar amounts using a single dollar sign like 24$ or enclose math between two dollar signs: $a^2 + b^2 = c^2$.

Display math uses double dollars:

int_0^1 x^2 , dx = frac{1}{3}

8.1 Theorem environments

Fenced divs with theorem-type classes get automatic numbering and can be cross-referenced.

Theorem 1. *In a right triangle, the square of the hypotenuse equals the sum of the squares of the other two sides:* $a^2 + b^2 = c^2$.

Proof. Let a right triangle have legs a, b and hypotenuse c. By constructing squares on each side and comparing areas, we obtain $a^2 + b^2 = c^2$. □

By Theorem 1, the relationship is fundamental to Euclidean geometry.

The supported theorem types are: `theorem`, `lemma`, `corollary`, `proposition`, `conjecture`, `definition`, `example`, `exercise`, `solution`, `remark`, and `algorithm`. Each type maintains its own counter.

Callouts

Callout divs highlight important information. Five types are available: `callout-note`, `callout-warning`, `callout-tip`, `callout-caution`, and `callout-important`. They can be nested.

> **ℹ Note**
> This is a note callout. Use it for supplementary information.
>
> > **⚠ Warning**
> > Callouts can be nested. This warning appears inside the note above.

Conditional content

Fenced divs with `.content-visible` or `.content-hidden` classes control which content appears in each output format.

This paragraph is hidden in LaTeX but visible in all other formats.

Format aliases are supported: `latex` and `pdf` both match LaTeX output, `markdown` and `md` both match Markdown.

Raw blocks

Raw blocks inject format-specific markup that passes through verbatim when the output format matches and is dropped otherwise.

Shortcodes

Shortcodes are inline directives written as `{{< name args >}}`. They are processed during rendering and replaced with format-aware output.

12.1 pagebreak
Insert a native page break. Produces format-appropriate output: `<div style="page-break-after: always;">` in HTML, `\newpage{}` in LaTeX, `#pagebreak()` in Typst.

Page one content.

Page two content.

12.2 meta
Print a value from the document's YAML front matter.

The title of this document is "*Calepin* Tutorial (PDF)".

This works with standard fields (`title`, `subtitle`, `author`, `date`, `abstract`) and custom metadata.

12.3 env
Print a system environment variable.

The `SHELL` environment variable on the local machine is: /bin/zsh.

If the variable is not set, the shortcode produces an empty string.

Citations

Calepin processes citations from BibTeX files listed in the YAML front matter under `bibliography`. Use `@key` for author-year citations and `[-@key]` for year-only.

As Arel-Bundock et al. (2026) show, political science research is underpowered. This finding has important implications 2026.

Tabsets

Tabsets organize content into switchable tabs. In HTML, readers click to switch tabs. In LaTeX and Typst, tabs render as regular sections.

14.1 R

```r
x <- c(1, 2, 3)
mean(x)
```

```
[1] 2
```

14.2 Python

```python
x = [1, 2, 3]
sum(x) / len(x)
```

14.3 Julia

```julia
x = [1, 2, 3]
sum(x) / length(x)
```

Tabsets with the same `group` attribute switch together:

14.4 R
R version of example 2.

14.5 Python
Python version of example 2.

14.6 R
R content stays synced with the tabset above.

14.7 Python
Python content stays synced too.

Line blocks

Line blocks preserve line breaks and leading spaces. Prefix each line with `|`.

The limerick packs laughs anatomical
Into space that is quite economical.
  But the good ones I've seen
  So seldom are clean
And the clean ones so seldom are comical.

Footnotes

There are two ways to create footnotes. Named references use `[^label]` at the point of reference and `[^label]: text` for the definition elsewhere in the document. Inline footnotes use `^[text]` directly without a separate definition.

Named footnote[3] and inline footnote[4] .

Numbered sections

Add `number-sections: true` to the YAML front matter to automatically number all section headings.

The `.hidden` div

The `.hidden` div executes its content but produces no output.

The value computed silently: __CALEPIN_1f34_189e69e921651050___ERROR:object ‘secret’ not found.

References

Vincent Arel-Bundock, Ryan C Briggs, Hristos Doucouliagos, Marco M Aviña, Tom D Stanley. 2026. Quantitative political science research is greatly underpowered. The Journal of politics

[3]This is defined separately.
[4]This is defined right here.

Appendix

20.1 Reuse

Creative Commons Attribution 4.0

20.2 Citation

Vincent Arel-Bundock, Norah Jones. "**Calepin** Tutorial (PDF)". *Journal of Reproducible Research.* 1(1). (2026-03). DOI: 10.1234/calepin.2026

20.3 Copyright

Copyright 2026 Vincent Arel-Bundock

20.4 Funding

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